

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

May 2012

EC 202 Network Theory

Date: 02.05.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

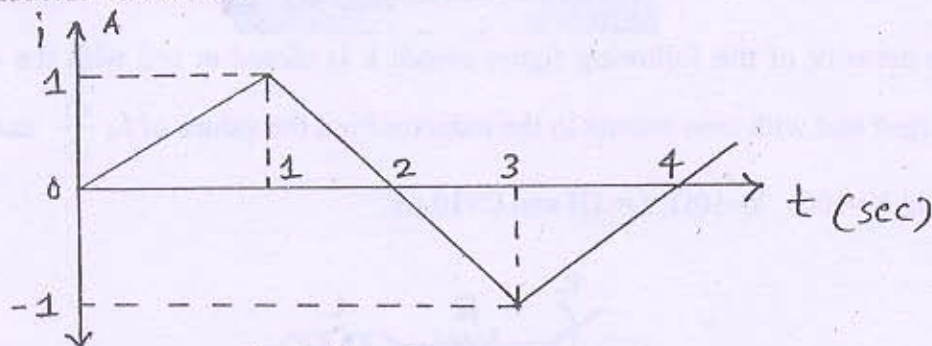
Maximum Marks: 70

Instructions:

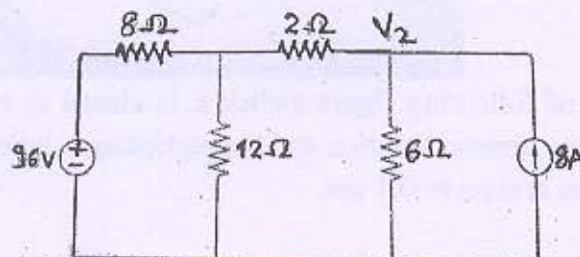
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain classification of networks. [05]
(b) Define Kirchhoff's voltage and current law. [02]
- Q - 2 (a) A current source, having waveform shown in following figure is connected across the terminals of an inductor of 10 mH. Show [05]
1) the voltage waveform and
2) the charge waveform through the device. Assume the initial conditions through the inductor to be zero.

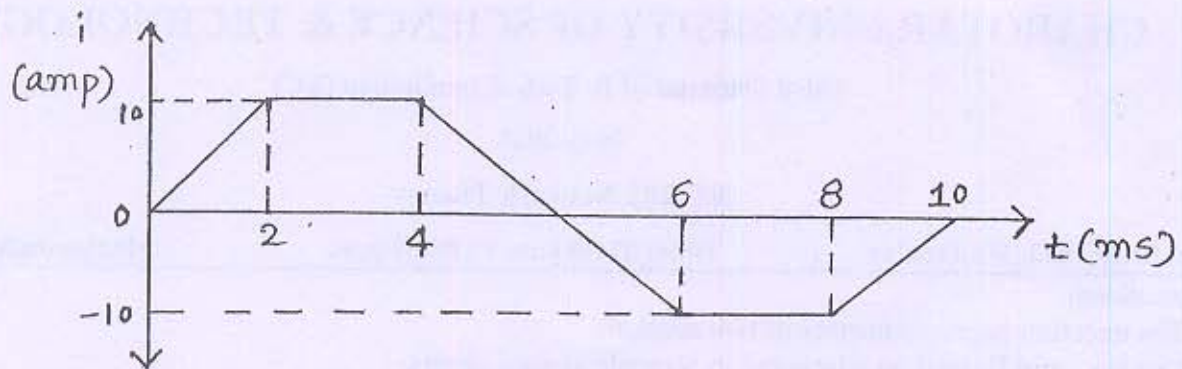


- (b) Explain source transformation rules. [05]
(c) Determine the voltage across 6Ω resistor in the network of the following figure using node analysis [04]

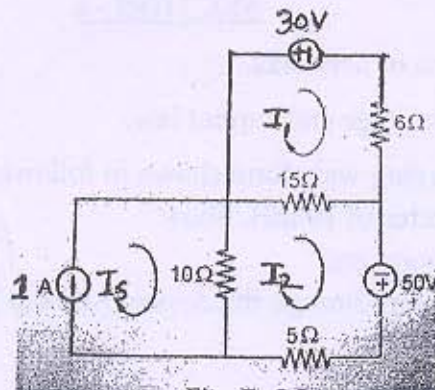


OR

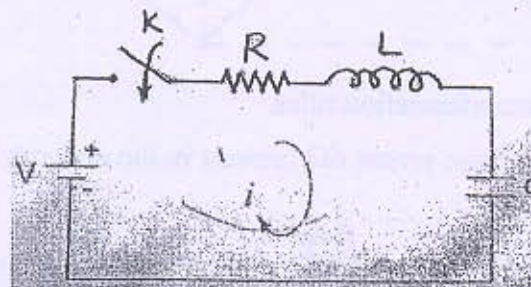
- Q - 2 (a) A single pure inductance of 3 mH passes a current of the waveform shown in following figure. Determine and sketch the voltage $v(t)$ and the instantaneous power $p(t)$. [05]



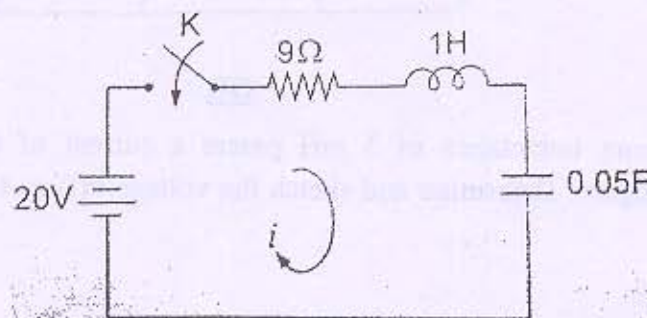
- (b) Explain the concept of duality in detail. [05]
- (c) Determine the mesh current i_1 and i_2 in the network shown below using mesh analysis. [04]



- Q-3 (a) In the network of the following figure switch k is closed at $t=0$ with the capacitor uncharged and with zero current in the inductor. Find the values of i , $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at $t=0^+$. if $V=100\text{V}$, $R=10\Omega$, $L=1\text{H}$ and $C=10\mu\text{F}$. [05]



- (b) In the circuit of following figure switch k is closed at $t=0$. For the element values given, obtain the general solution and the particular solution for current $i(t)$. Obtain the value of current at time $t=0.1\text{ sec}$. [05]

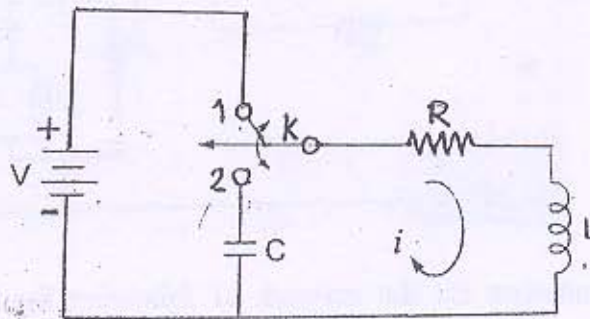


(c) Find the Laplace transform of $\sin \omega t$.

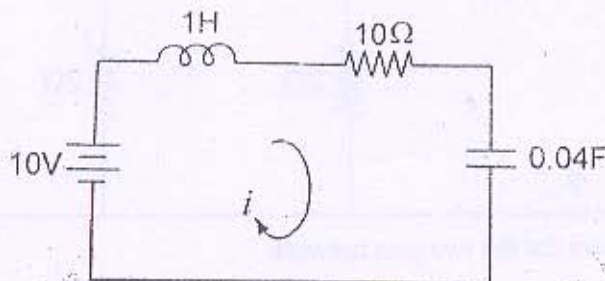
[04]

OR

- Q - 3 (a) In the network of following figure, the switch k is changed from position 1 to 2 at $t=0$. Find values of i , $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at $t=0^+$ if $R=1000\Omega$, $L=1H$, $C=0.1\mu F$ and $V=100V$. [05]

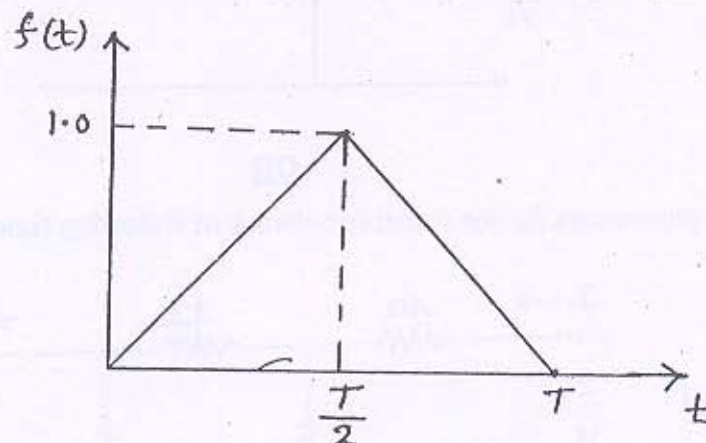


- (b) In the circuit of following figure, a DC voltage of 10 volts is suddenly applied to a series circuit consisting of resistor $R=10\Omega$, inductor $L=1H$ and capacitor $C=0.04F$. The capacitor is initially uncharged. Obtain the particular solution for current $i(t)$ in the circuit. [05]



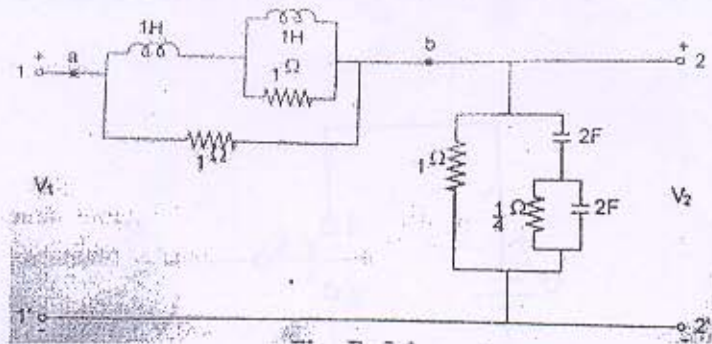
- (c) Find the Laplace transform of triangular pulse shown in figure.

[04]

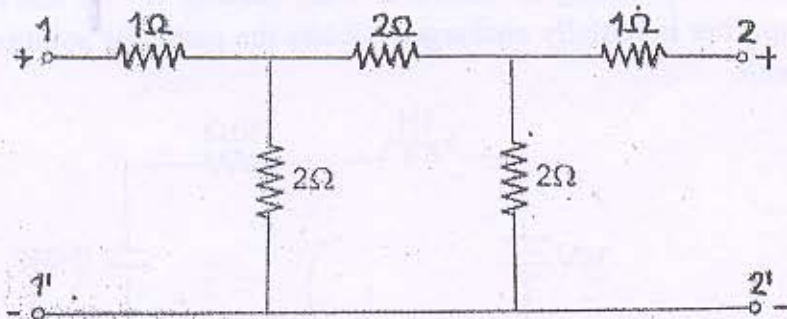


SECTION - II

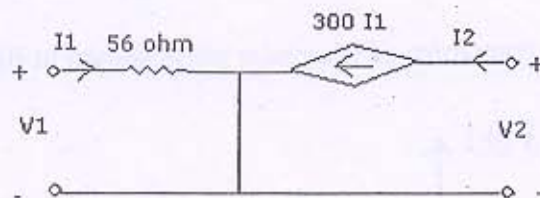
- Q - 4 For the network of following figure, show that with port 2 open, the input impedance at the port 1 is 1Ω . Also find the voltage ratio transfer function $G_{12}(s)$ for the two port network. [07]



- Q - 5 (a) Obtain Z parameters for the network of following figure. Draw the Z-parameter equivalent model and find whether the network is (a) reciprocal (b) symmetrical. [07]

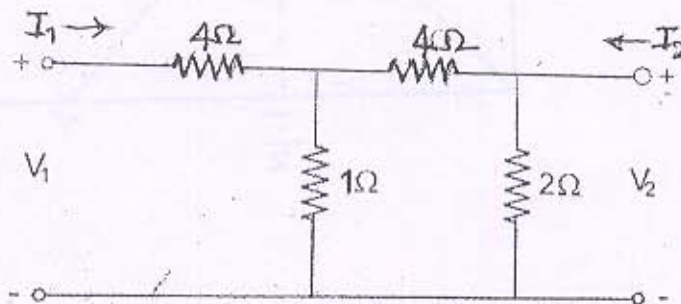


- (b) Find h-parameters for the two port network [07]



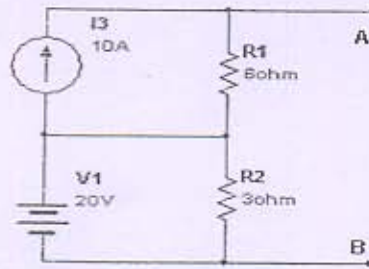
OR

- Q - 5 (a) Find the y-parameters for the resistive network of following figure. [07]



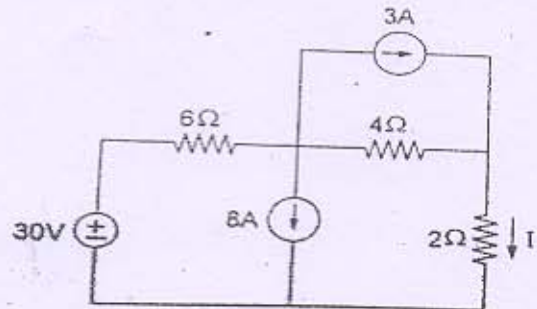
(b) Replace the given network shown in figure by Norton's equivalent network.

[07]



Q - 6 (a) Determine the value of i_1 using superposition in a network of following figure.

[07]



(b) Consider the RL series network in which the source is sinusoidal $v(t) = V \cos \omega t$.

[07]

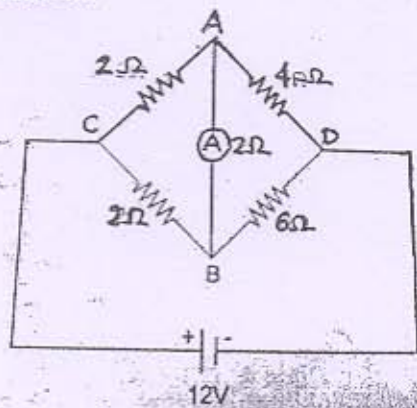
Find steady state solution for current $i(t)$.

hint: $V \cos \omega t = V \left(\frac{e^{j\omega t}}{2} + \frac{e^{-j\omega t}}{2} \right)$

OR

Q - 6 (a) Determine the current through the ammeter of resistance 2Ω connected in whistone bridge using Thevenin's theorem.

[07]



(b) For the network function $\frac{1}{1-j2\omega}$, where ω for both $\omega > 0$ and $\omega < 0$.

[07]

Sketch the following:

- Magnitude
- Phase
- Real part
- Imaginary part
